Laboratory 4: Brushless Permanent Magnet Motor Drives

Version 2.0

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Introduction

There is little doubt that in applications requiring low to medium power (<100kW) the brushless permanent magnet (PM) motor has received the most R&D attention over the last few decades given the technical and economical advances in rare earth magnets, power electronic converters and digital controllers.
In this laboratory session you will investigate the two principal motor types (brushless DC and permanent magnet AC) and their associated drives. The advantages of these machines can be summarised as follows:

- Brushless
- HIGH efficiency/power density
- Variable speed easily implemented
- Low torque ripple (Permanent Magnet AC)
- Simple controller (Brushless DC)

Industrial Servo Motors

Electric/Hybrid Vehicles

Aircraft Actuation

Aims and Objectives

Model 1: Brushless DC Motor Drive

- Back EMF v Commutation signal phasing
- Back EMF Constant K_e
- Torque v Current for Current Control Mode
- Torque Constant K_t
- Torque Ripple at Rated Current
- Torque v Speed curve at Rated Current

Model 2: Permanent Magnet AC Motor Drives

- Torque v Speed curve at Rated Current
- Torque Ripple at Rated Current
- Increasing high speed torque using Phase Advance

Brushless DC Motor Drive

In this experiment we want to investigate the implementation and operation of a Brushless DC motor drive. An existing model (Lab4_Model1.bak) for this is available on Moodle. Copy this onto the Desktop as outlined on the previous page and then open this model in Portunus. The model consists of 2 parts; the inverter/motor section as shown on Figure 1, and the control section as shown on Figure 2.

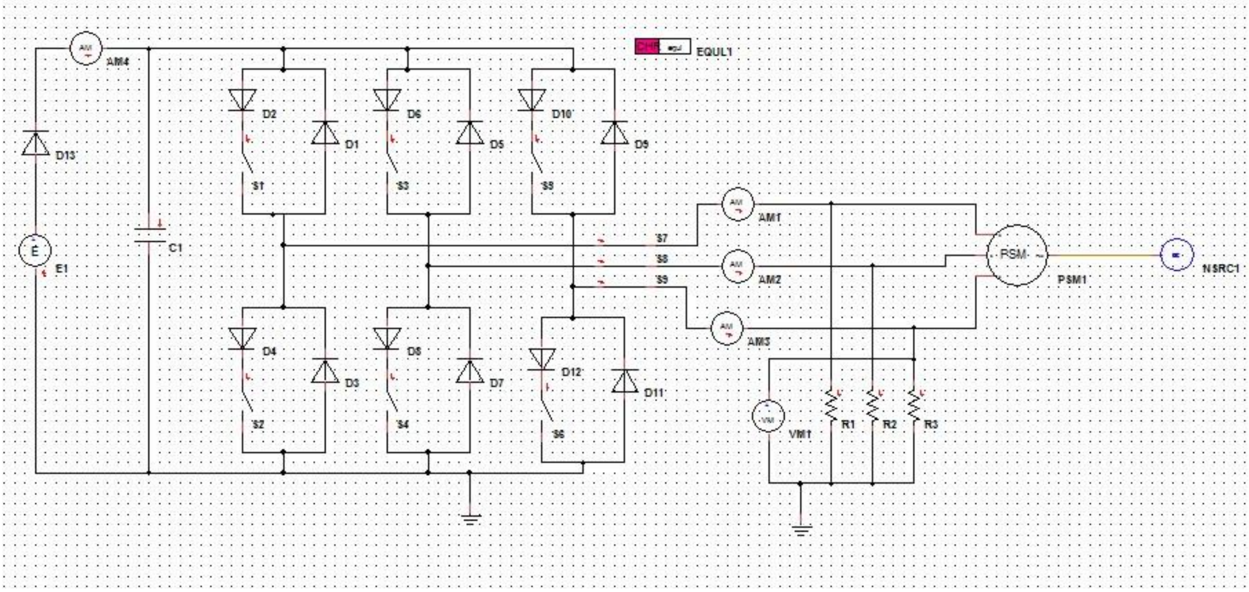


Figure 1: Brushless DC Power Inverter/Motor Model

The Inverter/Motor model consists of a DC voltage source (E1) providing power to a 3-phase inverter which connects to a 3 phase 2 Pole Brushless DC motor (PSM1). The motor is driven at constant speed using the speed source NSRC1.

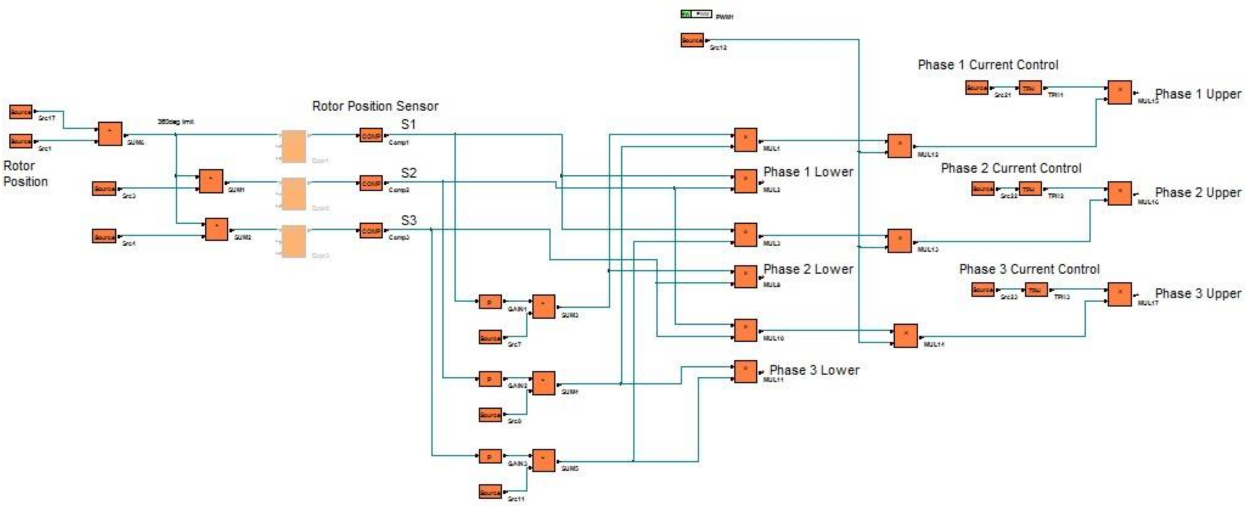


Figure 2: Brushless DC Controller

The Brushless DC controller uses rotor angle information Src1 (NSRC1.POS) to create the three commutation sector sensor signals S1, S2 and S3 which are subsequently decoded to create the commutation signals for phases 1-3 upper and lower gate drives. In addition, a 2kHz voltage PWM signal (Src12) and hysteresis current control (TPH1-3) are added to the control of each of the Upper phaseleg switches.

Exercise 1.0: Commutation Control

The first experiment will investigate the required phase relationship between the motor back EMFs and the switch commutation control signals. With respect to Figure 3, The back EMF for each the motor phases can be measured using a star connected resistor network (R1-R3) with the inverter disconnected from the motor which is achieved by opening the three switches S7-9 (Control signals = 0). The voltage across R1 measures Phase 1 Back EMF, R2 Phase 2 and R3 Phase 3. To display these voltages either connect the VM1 to the voltage you are interested in or simply select R1.V etc on the On Sheet display – it's your choice.

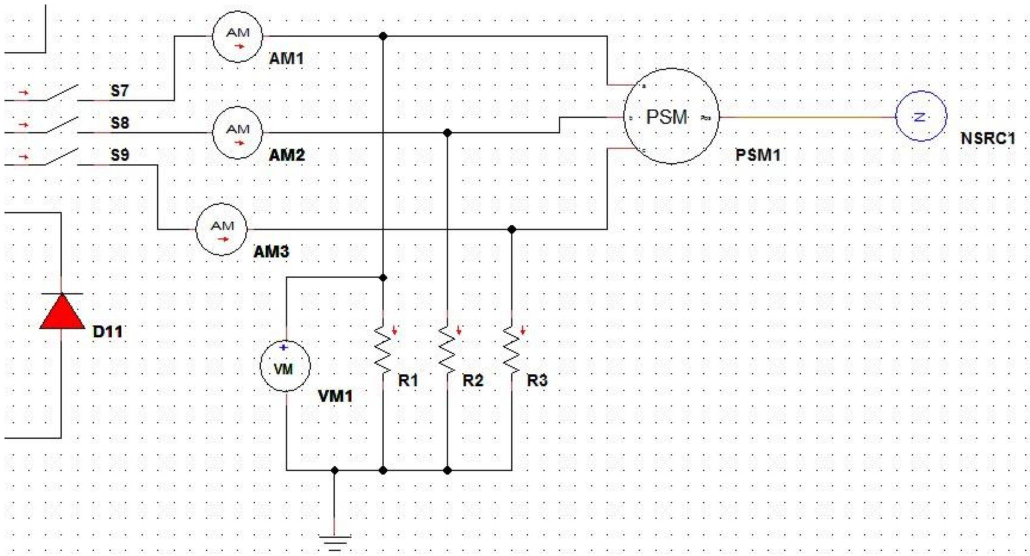
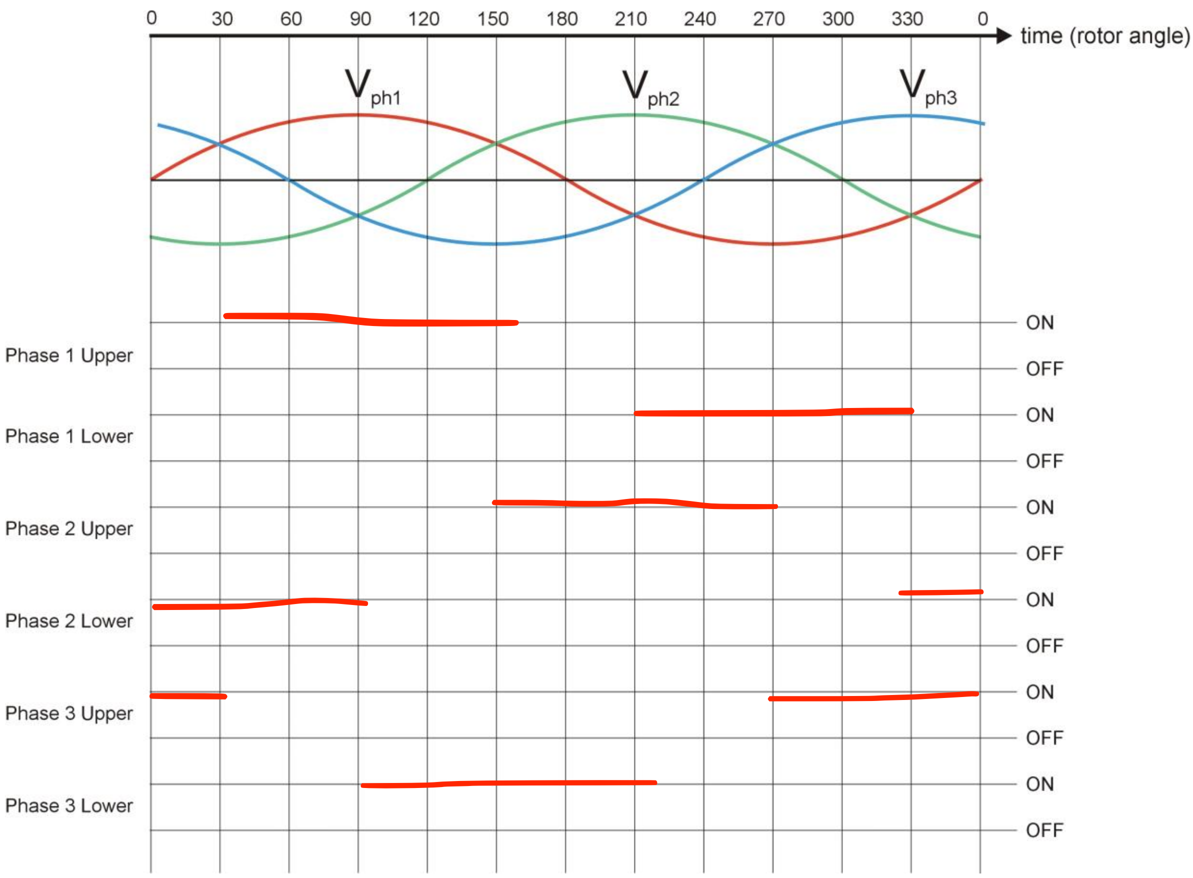


Figure 3: Back EMF Measurement

Procedure:

1. Add a ground connection to the star point of the resistor network R1-3 as shown on figure 3
2. Set the PWM Generator (PWM1) Duty = 1
3. Set the speed source NSRC1 = 1000rpm
4. Set TEND = 0.2s in the Simulation Control Panel [F9]
5. Send the relevant Back EMF (eg R1.V) and inverter switch control signals to an On Sheet display (I would suggest you do this one phase at a time!)

From this test or a series of tests complete the commutation timing diagram below (hint each switch is ON for 120°):



Also determine the Peak Back EMF at 1000rpm and from this determine the Motor Back EMF Constant Ke:

Peak Back EMF @ 1000rpm	31.4157	V
K_e	0.0314157	V/rpm
K_e (SI Units)	0.300	V / rad/s

Current (Hysteresis) Control Mode

We now want to do a series of tests running the machine in current control mode to determine the following:

1. Torque constant K_t
2. Torque Ripple
3. Torque v Speed Curve at rated current

The current controllers regulate the relevant phase current between two reference levels using a comparator with hysteresis. This is achieved by turning the upper gate drive signals on and off in response to the comparator output as shown on figure 4 (in a similar manner to how we controlled the braking resistor switch in the previous lab session). The upper current reference (I_{Upper}) is set by a variable (IRef) which can be accessed and changed using the **var** icon on the main toolbar (**next to the On Sheet Display Icon**). This value is then used as the upper threshold in each of the 3 TPH Comparator blocks in the controller section (check this to be sure). Also note that the lower current reference (I_{Lower}) in these TPH blocks is fixed at IRef – 0.5A, e.g. if IRef = 5A then I_{Upper} = 5A and I_{Lower} = 4.5A

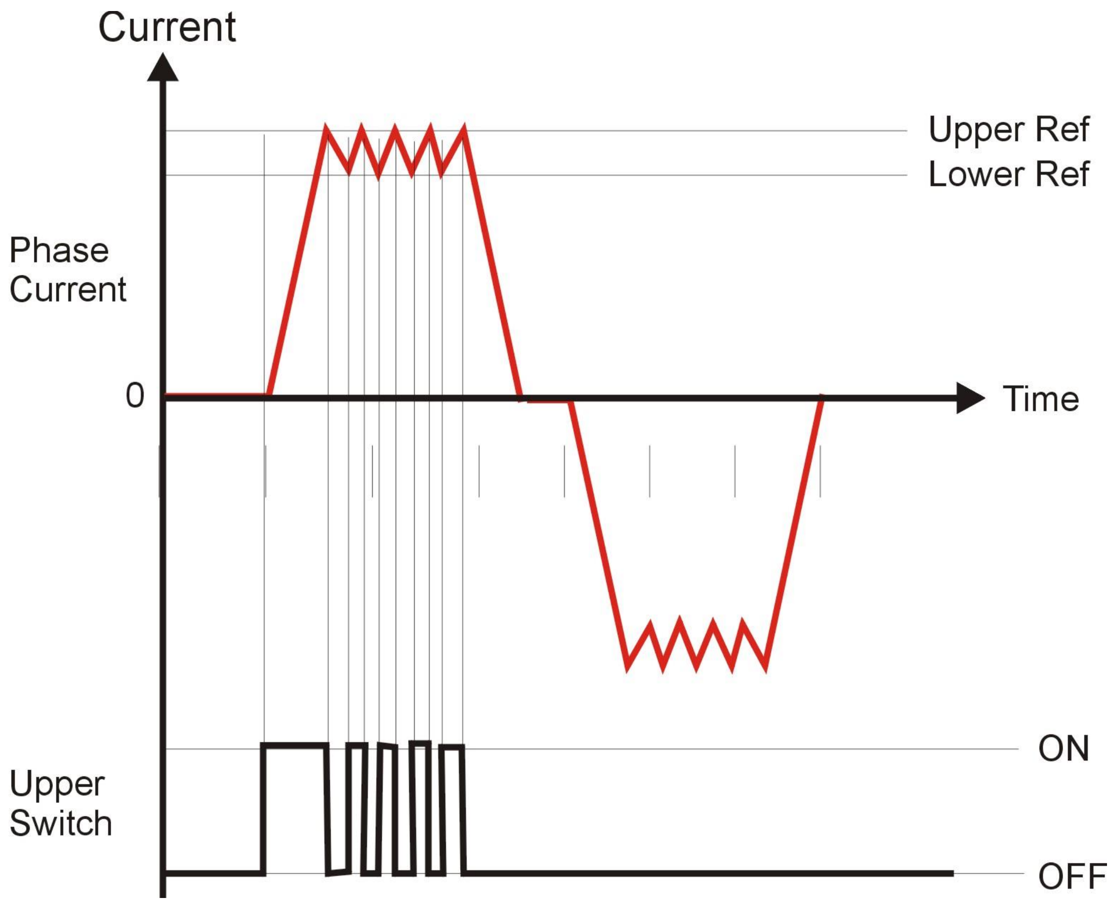


Figure 4: Hysteresis Current Control

We first need to reconnect the motor to the inverter as follows:

- Remove the earth connection from the resistor network star point.
- Set switches S7-9 control signal parameter = 1.

Exercise 1.1: Torque/Amp Constant

We need to measure and record the motor shaft torque, the necessary blocks to do this (as usual) being shown on Figure 5.

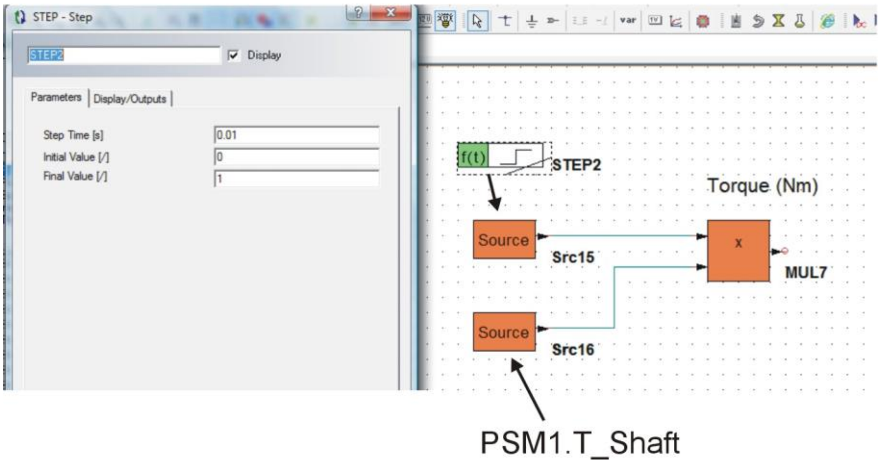


Figure 5: Shaft Torque Measurement

Motor Torque is directly available from the PSM1.T_SHAFT output parameter. However, this signal tends to have a VERY high transient value at the start of the simulation (not actually real just a simulation hiccup) so to remove this from the signal we multiply the torque waveform with a step function which is zero for the first 10ms of the simulation. The resultant signal MUL7.OUT should be sent to an On Sheet Display to determine motor torque (note it may NOT be MUL7 in your design!!). Note that the torque value from Portunus is negative for motoring (if you wish to make torque value positive then add a -1 gain before MUL7 block) I would also recommend that you output a motor phase current (AM1.I) on an On Sheet Display to verify that the machine is operating correctly (i.e. the current is controlled by Iref) at each test point

Set TEND = 0.2s in the Simulation Control Panel [F9]

Run a series of simulations at the following reference currents to determine the resultant torque:

Test Point	Reference Current (A)	Average Motor Torque (Nm)
1	4	1.88184
2	6	2.91107
3	8	3.97976
4	10	4.98404
5	12	6.00736

Note you will observe quite a bit of torque ripple for each of these. Using a Sana Block (with a sliding window set to 1 period of revolution) measure the average Torque value for each of the reference Currents. Graph Torque v Current and from this determine the Torque Constant Kt (SI Units):

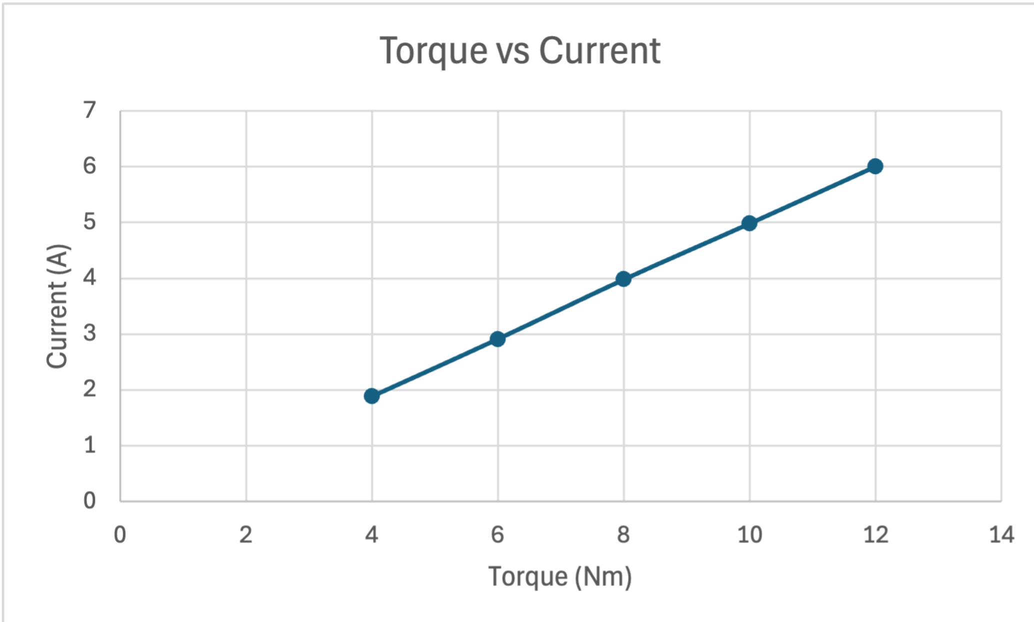
K_t	0.51477	Nm/A

Also, at the 12A Current Reference determine the % torque ripple (note that you can also use the Sana block to measure the Max and Min values!!):

$$Torque\_Ripple(\%) = \frac{\Delta Torque}{Torque_{AVG}} x100\%$$

$$Where:$$

$$\Delta Torque = Torque_{MAX} - Torque_{MIN}$$



Torque vs Current
Current (A)
Torque (Nm)
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14

Current Ref (A)	Max Torque (Nm)	Min Torque (Nm)	Torque Ripple (Nm)	Average Torque (Nm)	% Torque Ripple
12	6.89915	5.26199	1.63716	6.00736	27.2526%

We will consider 12A to be the rated current for this machine. The next test is to determine the Torque v Speed curve when operating at this reference current.

Exercise 1.2: Torque v Speed Curve at Rated Current

To determine this, we simply need to control the motor speed using the NSRC1 Speed Source and measure the resultant torque when operating at 12A rated current.

Set **TEND = 0.2s** in the Simulation Control Panel [F9]

Measure the motor torque at the following speeds and either measure or calculate the resultant output (mechanical) power:

Test Point	Motor Speed (rpm)	Peak Positive Motor Phase Current (A)	Motor Torque (Nm)	Output (Mech) Power (W)
1	500	12.0057	6.4162	335.9514
2	1000	12.0051	6.0081	629.1668
3	2000	12.002	5.5978	1172.4005
4	2500	10.0211	4.7753	1250.1706
5	2800	8.6956	3.4291	1005.4646
6	3200	4.8900	1.8665	625.4702

Graph Motor Torque v Speed and Output Power v Speed on separate graphs and from these estimate the following:

No Load Speed	3900	rpm
Max Power Point (speed)	1000	rpm

Using the 'classical' method to measure supply power determine the input DC electrical supply power and motor efficiency when operating at **2500rpm and a Current Reference (IRef) =12A**:

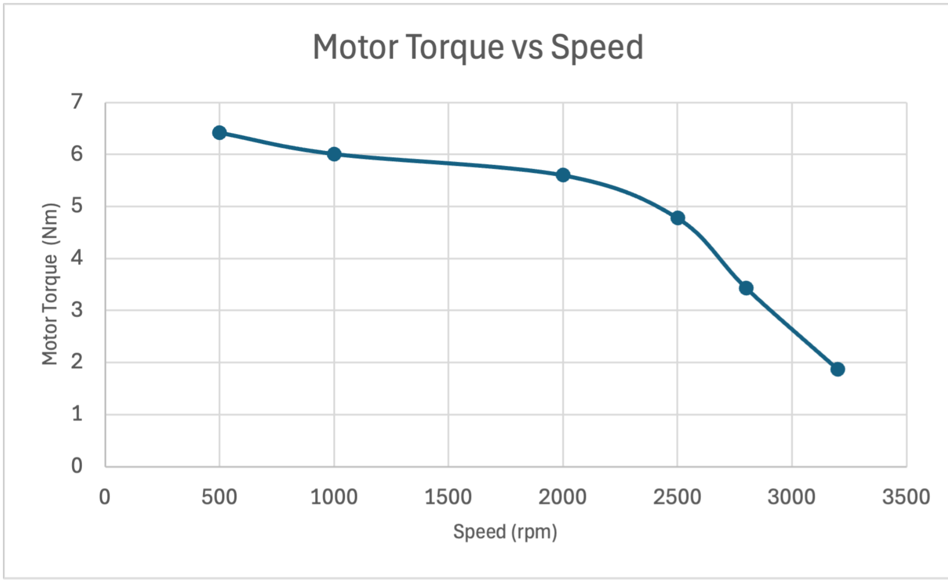
$$P_{ave} = \frac{1}{\tau}\int_0^{\tau} v_{dc}.i_{dc}.\partial t$$

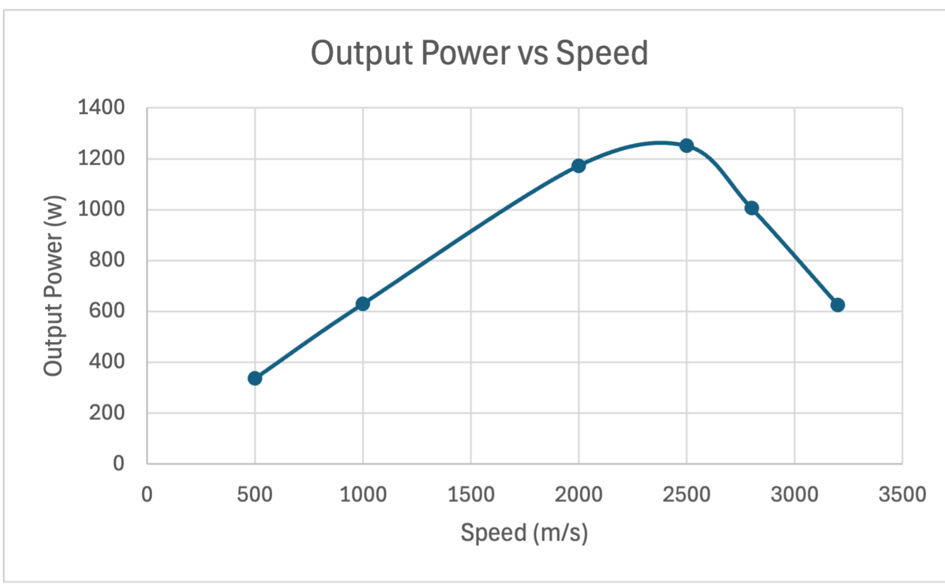
where:

vdc = instantaneous DC supply voltage
idc = instantaneous DC supply current
τ = one period (s) of rotation

Note: unfortunately, the DC Supply current waveform has a large spike at the start of the simulation so you will need to multiply E1.I with a step function (t_{step} = 0.01) to remove this before multiplying with E1.V

Period of rotation	0.024	s
Output (Mech) Power	1264.24	W
Input (Elec) Power	1400.65	W
Motor Efficiency	90.2610	%


Motor Torque vs Speed
Motor Torque (Nm)
7
6
5
4
3
2
1
0
0
500
1000
1500
2000
2500
3000
3500
Speed (rpm)


Output Power vs Speed
Output Power (w)
1400
1200
1000
800
600
400
200
0
0
500
1000
1500
2000
2500
3000
3500
Speed (m/s)

Model 2: Permanent Magnet AC Motor Drive

We will now move onto investigating the alternative permanent magnet AC motor drive. Once again, a model is available for this (Lab4_Model2.bak) which should be downloaded and saved as before. This model again consists of two parts: the inverter/motor section (which is identical to the brushless DC model) and the 3-phase sinusoidal PI current regulators shown on Figure 6.

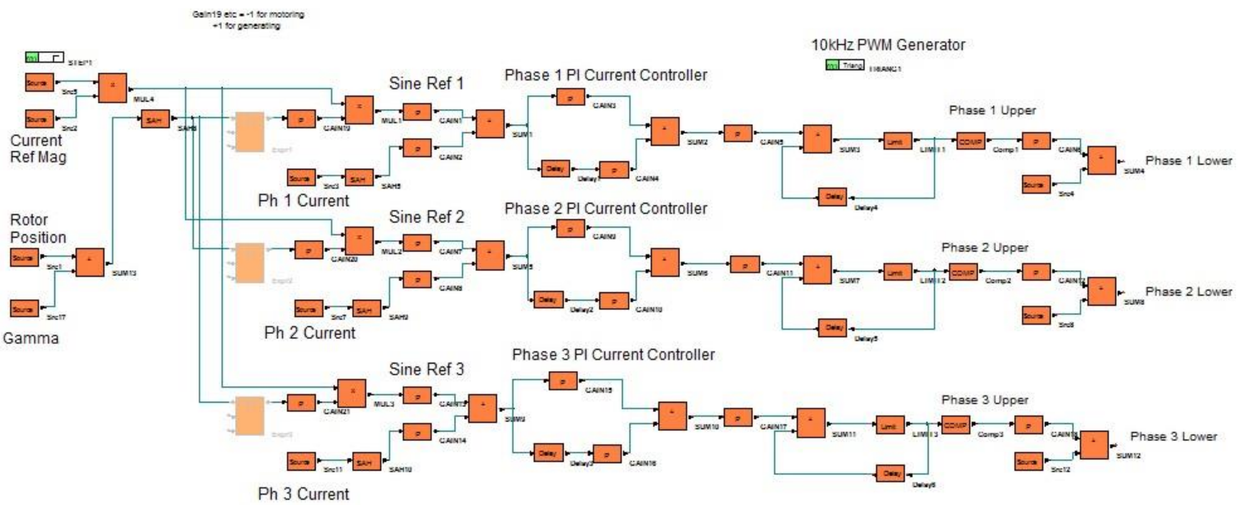


Figure 6: 3 Phase Sinusoidal PI Current Regulators

The current regulators align the 3 sinusoidal current references in phase with the respective phase back EMF waveform and then regulate the actual motor currents to track the associated reference, the result being (hopefully!) sinusoidal phase currents.

In this next section we will investigate the effects of Phase Advance on the Torque v Speed curve, the aim being to increase the torque above baseline speed and therefore output higher power for the same machine. As can be seen on Figure 7 Phase Advance places the current reference waveforms ahead (leading) of their respective back EMF's the advantages being that the back EMF isn't quite so strong in this region therefore there is less voltage opposing the DC Supply and so the phase current can reach its reference levels. (there are other advantages as well due to reluctance torque in salient pole machines but we haven't really went into this so I won't mention it again!).

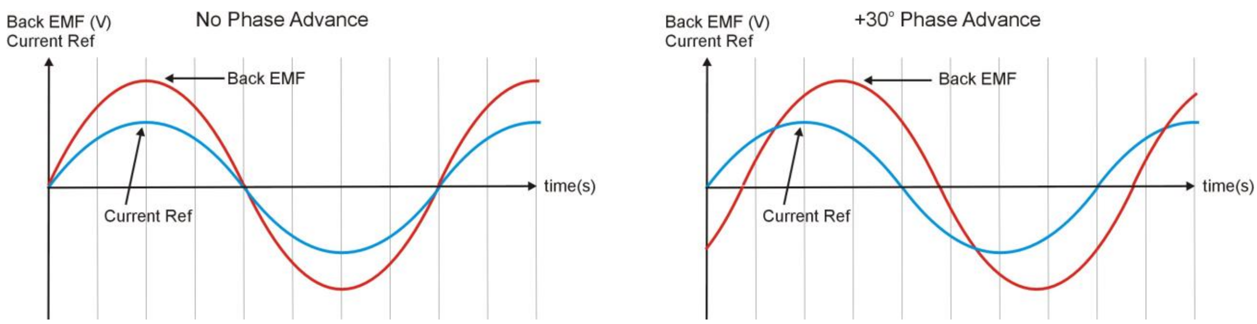


Figure 7: Phase Advance

(note: 30° phase advance is just an example, Gamma value can be any angle)

Exercise 2.1: Torque v Speed curve with Phase Advance = 0°

We want to repeat the torque v speed tests carried out for the Brushless DC drive but this time we want to include the measurement of input power from the DC power supply such that we can also determine machine efficiency. The generic term for the measurement of average (real) power is again the way to go:

$$P_{ave} = \frac{1}{\tau}\int_0^\tau v_{dc}.i_{dc}.\partial t$$

where:

V_{dc} = instantaneous DC supply voltage
i_{dc} = instantaneous DC supply current
τ = one period of motor phase current

$$Speed(rpm) = \frac{f_s.120}{P}$$

where:

number of poles P = 2 for this machine.

$$\tau = \frac{1}{f_s}$$

Note: again, unfortunately the DC Supply current waveform has a massive spike at the start of the simulation so you will need to multiply E1.I with a step function (t_{step} = 0.01) before multiplying with E1.V

The necessary blocks to calculate output motor torque and output power (as before) should also be included in this model.

Also output a motor phase current (eg AM1.I) to an On Sheet Display to check quality of the sinusoidal current controllers.

Set TEND = 0.2s in the Simulation Control Panel [F9]

Setting the Speed Source NSRC1 accordingly determine the torque v speed curve and associated operating parameters at a **reference current of 12A peak (set in Src2)** for the following motor speeds, noting the 'quality' of the phase current at each test point (i.e. how sinusoidal it is):

Test Point	Motor Speed (rpm)	Period of Motor Phase Current (ms)	Peak Motor Phase Current (A)	Motor Torque (Nm)	Input Power (W)	Output Power (W)	Efficiency (%)
1	500	120	12.5191	5.3513	442.931	280.1934	63.2589
2	1000	60	12.4306	5.3273	704.175	557.8736	79.2234
3	2500	24	12.2574	5.2799	1537.86	1382.2736	89.8830
4	3000	20	10.6369	4.5203	1533.58	1420.0941	92.5999
5	3200	18.75	9.1305	3.7698	1353.34	1263.2721	93.3448
6	3600	16.67	5.2809	1.8128	705.798	683.4095	96.8279
7	3800	15.79	3.0690	1.0328	424.385	410.9873	96.8430

Graph torque v speed for these results and indicate points where phase currents are no longer sinusoidal.

Also measure the torque ripple at 1000rpm point and comment on how it compares with the Brushless DC torque ripple at this speed and suggest an application where low torque ripple is important:

Current Ref (A)	Max Torque (Nm)	Min Torque (Nm)	Torque Ripple (Nm)	Average Torque (Nm)	% Torque Ripple
12	5.6011	4.9863	0.6148	5.3277	11.5397

Comment/Application:
At 1000 rpm the PMAC drive gives about 11.5% torque ripple, which is much lower than the Brushless DC machine at the same current (≈27%), so the PMAC produces a much smoother torque. Low torque ripple is important in applications where vibration and acoustic noise must be minimised, for example precision servo drives/robot joints or electric vehicle traction motors.

Suggest a simple equation based on Back EMF and phase current to calculate output (mech) power (hint – its similar to the equation used in the Brushed DC motor) and use this to calculate the output power at the 1000rpm operating point:

P_out = 3E_ph,rms * I_ph,rms
=1.5 * E_ph,pk * I_ph,pk

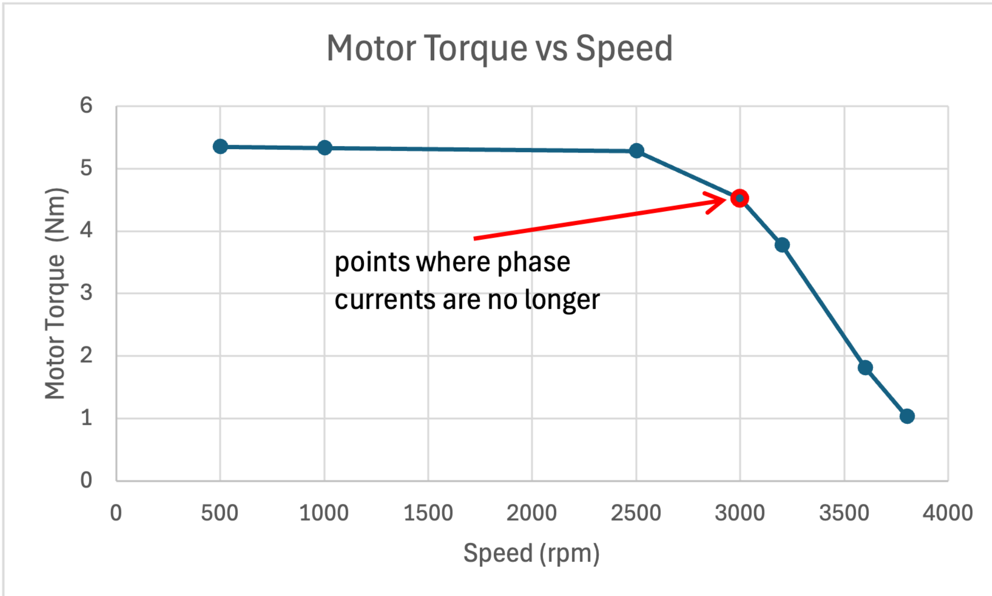
At 1000 rpm we have:
P_out = 1.5 * 31.42 * 12.43 = 0.59 kw
So the mechanical output power at 1000 rpm is ≈ 0.59 kW, which is close to the simulated value (~0.56 kW).

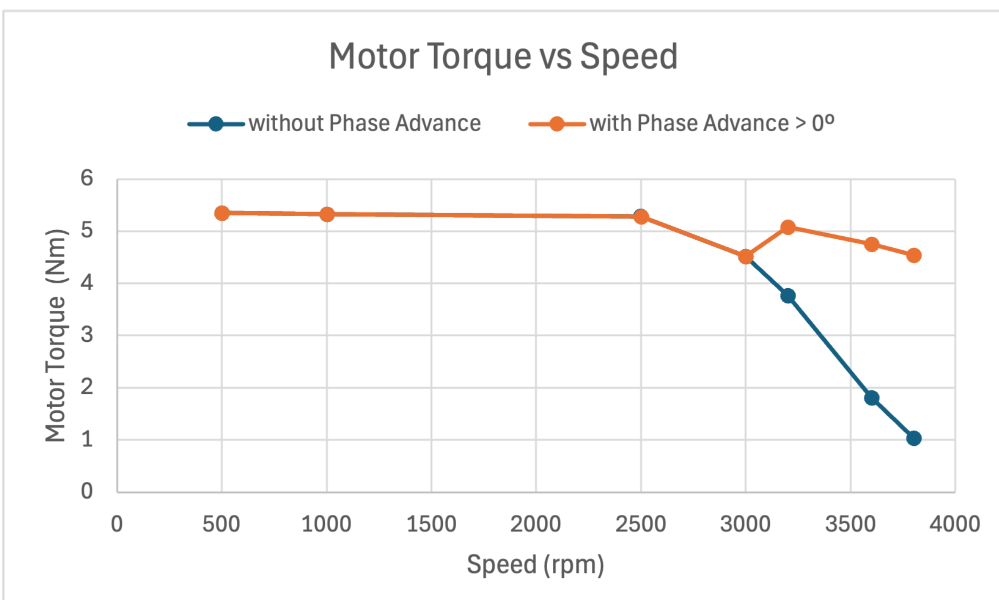
Exercise 2.2: Torque v Speed curve with Phase Advance > 0º

The challenge now is to determine empirically (fancy name for trial and error!) through simulation the value of Phase Advance for each of the speeds above baseline speed which results in maximum torque and therefore maximum power at each of these speeds. Suggested range for Phase Advance (Gamma) is 0 < Gamma < 40º and test at 5° resolution. Phase Advance (Gamma) is input in DEGREES (positive value for advance) in the Src17 Block (bottom left of model).

Test Point	Motor Speed (rpm)	Phase Advance (Deg)	Peak Motor Phase Current (A)	Motor Torque (Nm)	Output Power (W)
1	3200	15	11.9653	5.0755	1700.81
2	3600	25	11.9551	4.7505	1790.90
3	3800	30	11.9591	4.5380	1805.83

Add these points onto the Torque v Speed curve from Exercise 2.1


Motor Torque vs Speed
points where phase
currents are no longer
Motor Torque (Nm)
Speed (rpm)
0
1
2
3
4
5
6
0
500
1000
1500
2000
2500
3000
3500
4000


Motor Torque vs Speed
without Phase Advance
with Phase Advance > 0º
Motor Torque (Nm)
Speed (rpm)
0
1
2
3
4
5
6
0
500
1000
1500
2000
2500
3000
3500
4000

Appendix

Accessing Existing Models

In this laboratory session you are required to access an existing Portunus models (Lab4_Model1.bak and Lab4_Model2.bak) from Moodle page ENG4187. Copy these files into a known directory on your PC. To open this file in Portunus select ‘File-Open’ and go to the directory where you have stored the files. Note that in Portunus you should select ‘all files’ in the **Open File** panel to access the .bak file. If you subsequently want to save models then choose the relevant Lab4_Model’X’**.bak** file as it is MUCH smaller than the corresponding .ecd file but contain all the necessary information and can be subsequently opened from Portunus using the select ‘all files’ in the **Open File** panel.